StudyBuddy College Notes - Data Structures & Algorithms

Chapter 1: Asymptotic Analysis and Big-O

Asymptotic analysis allows us to measure the time and space complexity of an algorithm relative to input size n.
Big-O notation describes an upper bound on time complexity.
Common complexity classes:
- O(1): Constant time (e.g., array index lookup)
- O(log n): Logarithmic time (e.g., binary search)
- O(n): Linear time (e.g., linear scan)
- O(n log n): Linearithmic time (e.g., Merge Sort, QuickSort average)
- O(n^2): Quadratic time (e.g., Bubble Sort, Selection Sort)

Chapter 2: Trees and Graphs

A Binary Search Tree (BST) maintains the property that for every node X, all keys in its left subtree are strictly sm
Traversal types:
- In-order traversal visits nodes in ascending order.
- Pre-order traversal is used to clone trees.
- Post-order traversal is used for tree deletion.

Graph representations:
- Adjacency Matrix: Space complexity O(V^2), optimal for dense graphs.
- Adjacency List: Space complexity O(V + E), optimal for sparse graphs.